

THE QUESTION

Q.14 · DMG 2019

In a fold, rock salt has broken through the more brittle overlying rocks and occupied its core. The structure is known as

- a Disharmonic fold
- b Drag fold
- c Box fold
- d **Diapir fold**

D

CORRECT ANSWER

Diapir fold — salt that pierces bedding in overlying strata and occupies the core

TEXTBOOK DEFINITIONS — THE AUTHORITY

TABLE 2.5 — TERMINOLOGY OF SALT STRUCTURES · PAGE 28 · VAN DER PLUIJM & MARSHAK

Salt diapir: "A salt structure that **pierces bedding in overlying strata.**"

Salt dome: "An upward salt bulge relative to the source layer that is roughly symmetric in plan view. Strata overlying these structures conformably warp around the structure."

📖 Table 2.5, Page 28 — Chapter 2: Primary and Nontectonic Structures

§2.3 SALT STRUCTURES · PAGE 26 · VAN DER PLUIJM & MARSHAK

"Salt is a nonporous and essentially incompressible material... salt is much weaker and, as a consequence, is able to **flow like a viscous fluid** under conditions in which other sedimentary rocks behave in a brittle fashion... If gravity is the only reason for salt movement, the deformation resulting from its movement is called **halokinesis.**"

📖 Page 26 — §2.3

§2.3.2 GEOMETRY OF SALT STRUCTURES · PAGE 27-28

"If the salt rises after the overlying strata have already been deposited, then the rising salt will **warp and eventually break through the overlying strata.** This process is called **upbuilding**... When salt diapirs approach the surface, the overlying strata are arched up and therefore are locally stretched, resulting in the development of normal faults in a complex array over the crest of the salt structure."

📖 Pages 27-29 — §2.3.2

TABLE 2.5	TERMINOLOGY OF SALT STRUCTURES
Detached bulb	Stems or walls connecting salt stocks to the source bed may pinch out, so that salt that was originally separated by the stem or wall becomes juxtaposed. Flow of the salt into bulbs results in folds that wrap around themselves and thus have circular profiles.
Detached canopy	Stems or walls connecting salt canopies to the source bed may pinch out, so that salt that was originally separated by the stem or wall becomes juxtaposed.
Salt anticline	An upward salt bulge relative to the source layer that is elongate in plan view. Strata overlying these structures conformably warp around the structure.
Salt canopy	In regions where the source bed was quite thick so that many diapirs form, salt walls or salt stocks may spread out and merge at a higher stratigraphic level. Salt canopies may flow dominantly in one direction in response to gravity, and if so are sometimes called <i>salt-tongue canopies</i> .
Salt diapir	A salt structure that pierces bedding in overlying strata.
Salt dome	An upward salt bulge relative to the source layer that is roughly symmetric in plan view. Strata overlying these structures conformably warp around the structure.
Salt glacier	Salt that flows out over the land when the salt diapir pierces ground surface (Figure 2.17).
Salt pillow	Same as <i>salt dome</i> .
Salt roller	An asymmetric salt dome that resembles an ocean wave.

TABLE 2.5 — Terminology of Salt Structures. Note **Salt diapir**: "A salt structure that pierces bedding in overlying strata" — directly matching the question description. *Salt dome merely warps strata; the diapir breaks through.*

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Salt stock	Equant salt diapir in plan view. Mature salt diapirs are generally narrower at depth and broader at the top; the lower part is called the <i>stem</i> and the upper part is called the <i>bulb</i> .
Salt wall	Elongate salt diapir in plan view.

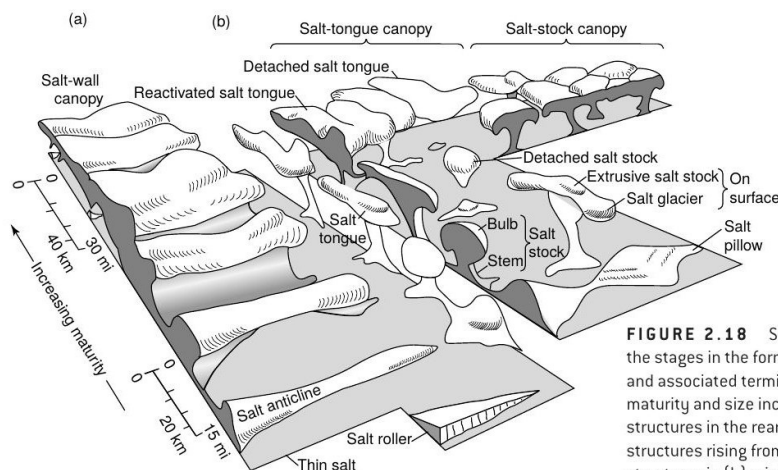


FIGURE 2.18 Schematic diagram showing the stages in the formation of salt structures and associated terminology. Structural maturity and size increase toward the structures in the rear. Sequence in (a) shows structures rising from line sources, whereas structures in (b) originate from point sources.

FIGURE 2.18 — Schematic diagram showing the stages in the formation of salt structures. Note the progression from salt anticline → salt roller → salt pillow → salt wall → salt stock (diapir). The diapir stage is where the salt **pierces through** overlying brittle strata.

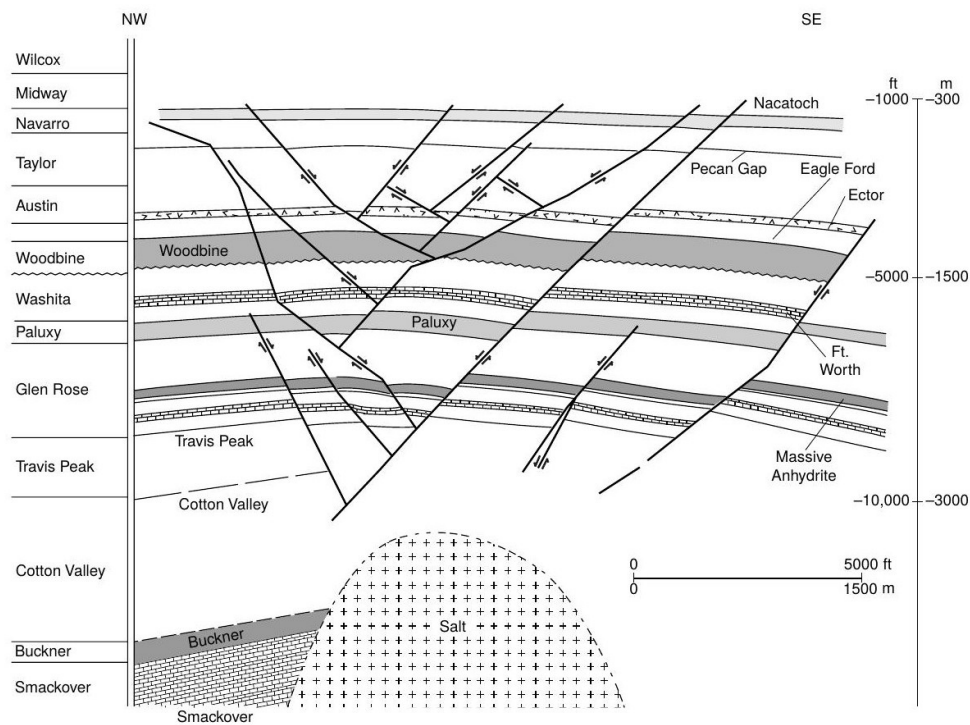


FIGURE 2.19 — Cross section illustrating a normal fault array over the top of a salt dome in Texas. Note the upward-piercing salt body (cross-hatched, center) that has broken through the overlying sedimentary strata — this is the diapir geometry described in the question. The overlying brittle strata are arched and faulted above.

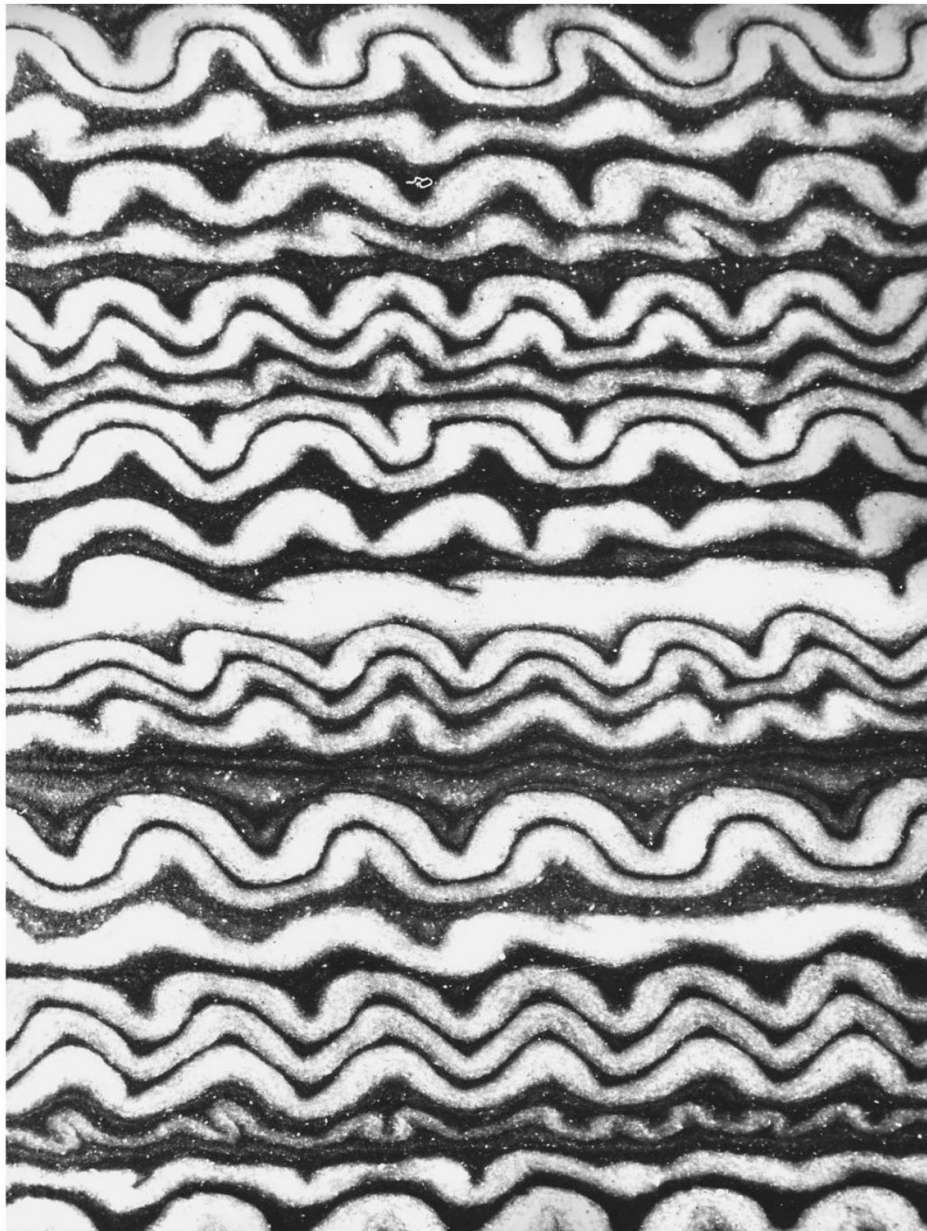


FIGURE 10.5 Small-scale disharmonic folds in anhydrite of the Permian Castile Formation in the Delaware Basin of Texas. White layers are anhydrite; dark layers consist of calcite that is rich in organic material (hence the dark color). Note that detachments occur in the organic-rich calcite layers and that the fold shapes (including box folds) in the anhydrite vary as a function of layer thickness.

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FIGURE 10.5 — Small-scale **disharmonic folds** in anhydrite of the Permian Castile Formation, Texas. White = anhydrite; dark = organic-rich calcite. Note that different layers fold with different wavelengths/amplitudes (disharmonic). Box folds are also visible — layer-detached, flat-topped geometries. This directly illustrates options (a) and (c).



FIGURE 10.18 Monocline in the Big Horn Mountains (Wyoming, USA).

10.5 SOME SPECIAL FOLD GEOMETRIES

We end the descriptive part of this chapter with a few special fold geometries that you may encounter in your field career. **Monoclines** are fold structures with only one tilted limb; the beds on either side of the tilted limb are horizontal. Monoclines typically result from a vertical offset in the subsurface near the tilted portion of the structure. The fault uplifts a block of relatively rigid igneous or metamorphic rock, and the overlying sedimentary layers drape over the edge of the uplifted block to form the monocline (Figure 10.18). Spectacular examples are found in the Colorado Plateau of the western United States. **Kink folds** are small folds (less than a meter) that are characterized by straight limbs and sharp hinges. Typically they occur in finely laminated (that is, strongly anisotropic) rocks, such as shales and slates (Figure 10.19). Sharply bending a deck of cards is a good analogy for the kinking process, because kink folds are formed by displacements between individual laminae (individual cards in the analogy). **Chevron folds** (Figure 10.8) are the larger-scale equivalent of kink folds.

The term **box fold** describes a geometry that is pretty self-explanatory (Figure 10.5). In order for a box fold to form, a layer must be detached from the underlying and overlying layers. They are therefore common in areas with weak basal layers, such as in the Jura Mountains of Switzerland. **Ptygmatic folds** are irregular and isolated



FIGURE 10.19 Kink folds in mica-rich portion of greywackes of the Cantabrian Mountains (northern Spain).

§10.5 Some Special Fold Geometries — Textbook page covering box folds, kink folds, monoclines, and ptygmatic folds. The **box fold** definition is here: "describes a geometry that is pretty self-explanatory... a layer must be detached from the underlying and overlying layers."

HOW A DIAPIR FORMS — STEP BY STEP

1 Salt Deposition in Passive-Margin Basin

Halite (NaCl) evaporites are deposited in rift basins during continental breakup. Thick salt layers accumulate at the base of the sedimentary sequence.

2 Burial and Density Inversion

Overlying sediments compact and become denser ($\approx 2500 \text{ kg/m}^3$). Salt stays at $\approx 2200 \text{ kg/m}^3$ and does not compact. At depths $>6 \text{ km}$, denser rock overlies less dense salt → **density inversion**.

3 Halokinesis Begins — Positive Buoyancy

The light salt is positively buoyant. Differential loading creates pressure gradients. Salt flows plastically upward from the **source layer** — the weak salt flows while surrounding rocks behave brittly.

4 Salt Dome / Pillow Stage

Salt bulges upward as a salt anticline or pillow. Overlying strata **conformably warp** around the structure without being pierced. This is the pre-diapir stage.

5 Diapir Stage — Piercing Through ✓ (THE ANSWER)

The buoyant salt breaks through the **more brittle overlying strata** and rises as a diapir (salt stock or salt wall). The salt now **occupies the core** of the structure. Normal faults develop above. This is exactly the scenario described in the question.

6 Salt Glacier (If Extruded to Surface)

If the diapir reaches the surface, salt flows out as a **salt glacier** (extrusive salt stock). Examples in western Iran.

ALL FOUR FOLD TYPES — CONCEPT CARDS

✓ **Diapir Fold**

A body of **mobile material (rock salt, magma, or overpressured shale)** that has pierced upward through more brittle overlying strata and occupies the core of the structure. Driven by buoyancy (density inversion) and differential loading. Classic example: salt diapirs / salt domes in the Gulf of Mexico.

✗ **Disharmonic Fold**

Folds in which different layers have **different wavelengths and/or amplitudes** — the fold pattern is not continuous from one layer to the next. Caused by differences in competency between layers. Nothing to do with salt piercing through strata.

✗ **Box Fold**

A fold with a **flat-topped hinge zone** (box-shaped in cross-section). Requires layer detachment from above and below. Common in weak basal layers (e.g., Jura Mountains). Purely a fold shape — no material piercing through anything.

✗ **Drag Fold**

An **outdated/discouraged term** for fault-related folds, previously thought to form by "drag" on fault surfaces. Textbook (p. 187): "We discourage and avoid the use of this term." Also = parasitic folds associated with larger folds. Completely unrelated to salt intrusion.

COMPARISON TABLE — ALL FOUR OPTIONS

FOLD TYPE	KEY FEATURE	MECHANISM	MATCHES QUESTION?
Diapir fold ✓	Mobile material pierces through brittle overlying strata; occupies core	Buoyancy + density inversion (halokinesis)	YES — exactly described
Disharmonic fold	Layers fold with different wavelengths/amplitudes	Competency contrast between layers	No — no piercing or core

Box fold	Flat-topped (box-shaped) fold hinge	Layer detachment + compression	No — just a fold shape
Drag fold	Abandoned term ; fault-related asymmetric fold	Parasitic folding near faults	No — discarded term, unrelated

OPTION-BY-OPTION ANALYSIS

a ✗ INCORRECT

"Disharmonic fold"

A disharmonic fold describes variation in fold geometry between layers — some layers fold with large amplitudes and short wavelengths, others with different dimensions. Van der Pluijm & Marshak (p. 241): "*If some layers have different wavelengths and/or amplitudes, the folds are disharmonic.*" Figure 10.5 shows disharmonic folds beautifully in alternating anhydrite and calcite layers. **No piercing, no core, no salt intrusion involved.**

→ Disharmonic = different fold shapes in different layers (Figure 10.5, p. 242)

b ✗ INCORRECT — DISCARDED TERM

"Drag fold"

The textbook actively discourages this term. Van der Pluijm & Marshak (p. 187): "*We close with a comment on the term drag fold, which is used to refer to all fault-related folds in older literature. This general application of the term is misleading... Thus we **discourage and avoid the use of this term.***" In another usage, "drag folds" = parasitic folds on the limbs of larger folds, previously misunderstood. Neither usage involves mobile material breaking through overlying rocks.

→ Drag fold = outdated/avoided term; related to faults or parasitic folding, NOT salt intrusion

c ✗ INCORRECT

"Box fold"

A box fold is purely a **shape descriptor** — the fold hinge is flat and wide, giving a box-like cross-section. Van der Pluijm & Marshak (p. 250): "*The term box fold describes a geometry that is pretty self-explanatory... In order for a box fold to form, a layer must be detached from the underlying and overlying layers.*" It is common in the Jura Mountains. The geometry is box-shaped in profile — it says nothing about any material piercing through other rocks.

→ Box fold = flat-topped fold shape caused by layer detachment (Figure 10.5 shows box folds in anhydrite)

d ✓ CORRECT ANSWER

"Diapir fold"

The question describes three key facts: (1) **rock salt** as the mobile material, (2) it has **broken through** the more brittle overlying rocks, and (3) it **occupies the core**. This is the textbook definition of a diapir. Table 2.5 (p. 28): "*Salt diapir: A salt structure that **pierces bedding in overlying strata.***" The process is halokinesis — buoyancy-driven upward flow of salt, which is ductile while surrounding rocks remain brittle. This is the fundamental contrast that drives piercement: salt flows, overlying rocks break.

→ Diapir = piercing mobile body; halokinesis = the process; "occupies the core" = classic diapir geometry

EXAM MEMORY AID

Diapir = Mobile material **PIERCES** through brittle rocks → occupies core

Disharmonic = Different wavelengths in different layers (no piercing)

Box fold = Flat-topped hinge shape (detached layers, Jura Mountains)

Drag fold = Discouraged/abandoned term (avoid in exams!)

Key words in question → "broken through" + "occupied its core" = **DIAPIR**

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

PROBABLE Q · 1

The gravity-driven movement of salt, independent of tectonic forces, is called

- (a) Diapirism
- (b) Halokinesis
- (c) Halotectonics
- (d) Salt tectonics

✓ **Answer: (b) — Van der Pluijm & Marshak, §2.3, p. 26**

PROBABLE Q · 2

A salt diapir differs from a salt dome in that the salt diapir

- (a) Is symmetric in plan view and warps overlying strata
- (b) Pierces the bedding in overlying strata
- (c) Is driven by tectonic compression only
- (d) Only forms in continental rift zones

✓ **Answer: (b) — Table 2.5, p. 28 — salt dome warps; salt diapir PIERCES**

PROBABLE Q · 3

Disharmonic folds develop when adjacent layers have

- (a) The same wavelength but different amplitudes only
- (b) Different wavelengths and/or amplitudes due to competency contrasts
- (c) Been pierced by a mobile core
- (d) Been detached from basal layers

✓ **Answer: (b) — Van der Pluijm & Marshak, p. 241**

PROBABLE Q · 4

Box folds are commonly developed in which of the following settings?

- (a) Deep crustal ductile shear zones
- (b) Regions with weak basal layers allowing layer detachment
- (c) Salt diapir regions where piercement has occurred
- (d) Transform fault zones

✓ **Answer: (b) — Van der Pluijm & Marshak, §10.5, p. 250 — Jura Mountains example**

PROBABLE Q · 5

The primary reason rock salt rises diapirically through overlying denser sedimentary rocks is

- (a) Tectonic compression squeezing salt upward
- (b) Positive buoyancy due to density inversion
- (c) Thermal expansion of salt at depth
- (d) Dehydration reactions increasing salt volume

✓ **Answer: (b)** — §2.3.1, p. 26-27; salt $\approx 2200 \text{ kg/m}^3$, surrounding rocks $\approx 2500 \text{ kg/m}^3$

PROBABLE Q · 6

When a salt diapir pierces through to the surface and flows laterally, the resulting structure is called

- (a) Salt wall
- (b) Salt pillow
- (c) Salt glacier
- (d) Detached canopy

✓ **Answer: (c)** — Table 2.5, p. 28; classic examples in western Iran

PROBABLE Q · 7

Which of the following geological settings most commonly hosts salt diapirs?

- (a) Active collision orogens
- (b) Passive continental margins with thick evaporite sequences
- (c) Mid-ocean ridge systems
- (d) Continental rift shoulders

✓ **Answer: (b)** — §2.3, p. 26; passive-margin basins discussed extensively